Ahmed Galal Elzeky

AI Engineer | LLMs | RAG | Agentic Systems | Arabic NLP

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Egyptian national | military service: exempted

Microsoft Certified Azure AI Engineer shipping enterprise generative AI end to end: a 22-microservice multi-tenant assistant in production, provider-agnostic LLMs with failover, and measured RAG. Published in Arabic LLM reasoning.

Work Experience

AI Engineer **Feb 2025 – Present**

Intelligent Systems (ISYS), IBM partner *Smart Village, Egypt*

SIA, enterprise multi-tenant AI assistant: 22 microservices, 470K lines of code, 4,800+ automated tests, 7-engineer team

- **Built SIA, live for about 70 staff and sized for a 1,000-seat client rollout**: 22 FastAPI microservices and Next.js, per-tenant databases and encryption keys, SSO (OIDC/SAML), SCIM, MFA, GDPR export and erasure, and bilingual English/Arabic with full RTL and Arabic-dialect speech.
- **Removed vendor lock-in from the LLM layer**, putting IBM watsonx (Llama 4), Anthropic Claude and OpenAI behind one interface with automatic failover, then adding a task-aware router across 20 task types, measured against production prompts, with cost stamped on each call.
- **Raised answer-confidence scoring from AUROC 0.35, worse than chance, to 0.994**, and grounded chat in a pgvector knowledge base with per-department access control so answers cite the source page; none of 1,614 answers carried a citation before.
- **Shipped 45 department-gated agent tools** (email, calendar, Word and Excel authoring, document comparison, web search) and an **AI recruitment pipeline** doing CV parsing, candidate matching, deduplication and automatic re-queue after an outage.
- **Trained Vodafone's engineering team** on IBM watsonx.ai over a 9-hour course (prompt engineering, tuning, LoRA/QLoRA, synthetic data, evaluation, RAG), and pitched SIA alongside watsonx.ai and watsonx Orchestrate to **C-suite audiences at ISYS conferences**.

Freelance AI & Full-Stack Engineer

Selected client and product work, 2025–2026 *Remote*

- **Shipped comicstop.co, a live bilingual storefront** *(2026)* on Next.js and Cloudflare Workers with headless Shopify, D1 and R2, where checkout revalidates every variant with Shopify so stale catalogue data cannot authorise a purchase; webhook-invalidated catalogue cache, PKCE sessions, Playwright suites.
- **Built Lango's comprehensible-input engine** *(2025–2026)*, generating stories at 90% known / 10% new vocabulary and validating that ratio after generation rather than trusting the model, over a self-hosted Fly.io NLP service (Japanese spaCy + SudachiPy, Chinese, Korean G2P); offline-first React Native app.
- **Automated weekly CFO reporting for Swipelabs** *(2025)* over the QuickBooks API (revenue, burn rate, overdue invoices, natural-language forecasting), plus a Slack assistant wired to Trello, ClickUp and Google Calendar.

- **AI Developer Intern**, Environ Adapt *(Aug 2024 – Oct 2024)*. Siamese network for image similarity replacing manual bale tagging, plus an image quality-control model.
- **AI Developer Intern**, ESLSCA University *(Feb 2024 – Apr 2024)*. GPT-based admissions assistant automating applicant responses and document checks.
- **Software Developer Intern**, IIG-HEC *(Jun 2023 – Dec 2023)*. Company site and a real-time blog publishing system on React and Firebase.

Publications & Awards

- **Published** "Multi-Hop Arabic LLM Reasoning in Complex QA", *Procedia Computer Science* 244 (2024) 66–75, Elsevier; third of five authors. Presented at ACLing 2024 (Dubai); ESLSCA research award.
- **Winner, GenAI Hackathon** hosted by Stakpak. Built in 48 hours the prototype that became Lango.
- **Times Higher Education Awards Arab World 2024**. Technological/Digital Innovation of the Year (Dubai); contributed to the winning university AI system.
- **Top-3 finalist, GESAwards Africa 2024** representing Egypt and North Africa with GPTube, a graduation project turning YouTube videos into quizzes and summaries with LangChain and OpenAI.
- **Tech content creator** with 37K followers, 484K likes and a 917K-view video, building an Arabic-speaking tech audience.

Skills

AI & LLM: OpenAI, Anthropic, Gemini and IBM watsonx APIs, LangChain, RAG, pgvector, agentic tool use, prompt engineering and tuning, LLM evaluation

Programming: Python, TypeScript/JavaScript, C#, SQL

Backend & Data: FastAPI, Django, NestJS, microservices, PostgreSQL, MongoDB, Redis

Frontend: React, Next.js, React Native/Expo, Tailwind

Cloud & DevOps: Azure, IBM watsonx, Google Cloud, Docker, Jenkins CI/CD, Cloudflare Workers, Supabase, Git

Certifications

Microsoft Certified: Azure AI Engineer Associate (earned May 2025, renewed). Credential ID FA9AAA1E4EAF1DC

IBM: watsonx.ai, watsonx Orchestrate and watsonx Assistant Practitioner Advanced (2025), plus 21 further IBM badges

Education

B.Sc. Computer Science: ESLSCA University, Cairo. GPA 3.5/4.0. Sep 2021 – Sep 2024

Languages: English fluent (IELTS 7.5). German A1 (learning). Japanese JLPT N4. Arabic native